Quantitative Macro Theory **HW #3**
Spring 2018 **Due at 5pm on Friday 2/9/2018**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (50 points) Let's calibrate the labor endowment shocks of a standard incomplete market (SIM) model to replicate the typical dynamics of individual earnings in the U.S. economy. A reasonable approximation to individual earnings dynamics is an AR(1) process like

$$\ln y_t = \rho \ln y_{t-1} + \varepsilon_t \text{ with } \varepsilon_t \ N(0, \sigma_\varepsilon),$$

where the autocorrelation coefficient is $\rho = 0.95$ and the standard deviation of the shocks is $\sigma_\varepsilon = 0.20$ at an annual frequency. (Recall that more sophisticated estimates include a transitory component to capture measurement error, as well as less persistent shocks, and a fixed individual component to capture the effect of education, ability, etc.)

   (a) (10 points) Use the Tauchen's method discussed in class to approximate the AR(1) described above as a Markov chain. Set the number of grid points $m = 5$ and $\lambda = 2$. Display the values of y and the transition matrix, P. Note that you will approximate $z = \ln y$ so you need to transform it, $y = \exp z$.

   (b) (10 points) Set the initial distribution as a uniform over the five points, $M_0 = [1/5, 1/5, 1/5, 1/5, 1/5]$. Given the transition P obtained in (a), compute the next period distribution M_1.

   (c) (10 points) Compute the stationary distribution; i.e., M^* such that

$$P \times M^* = M^*.$$

   (d) (10 points) Using the stationary distribution, compute and present in a table the value of the following statistics for y: $var(log(y))$, coefficient of variation, Gini coefficient, mean / median, percentile 90 / percentile 50, percentile 50 / percentile 10. Using the stationary distribution, plot a histogram of y and a Lorenz curve of y.

   (e) (10 points) Redo the previous question but setting the number of points for y at 50; i.e., $m = 50$. Put the result as a different column in the same table.

2. (50 points) Go to Federal Reserve website for the 2016 Survey of Consumer Finances and obtain the file "rscfp2016.dta", which is the Stata **extract** data. Note that in that file you won't have all the variable (the full data set is p16i6.dta) but this one is friendlier to work than the full data set. (Note: remember to use weights for all your computations).

(a) (10 points) Prepare a table that presents the values you found in the previous question for the statistics in the case $m = 50$ with the same values computed in the SCF 2016.

(b) (10 points) Plot in a figure the histogram for income in the SCF 2016 together with the one you obtained in question (1) for of y. Do the same for the Lorenz curve.

(c) (10 points) Compute and present in a table the value of the following statistics for income in the SCF 2016 by level of education of the household head (No high school diploma, High school diploma, Some college, and College degree): mean(income), median(income), var(log(income)), coefficient of variation income, Gini coefficient of income.

(d) (20 points) Obtain from here all the previous SCFs since 1989 (excluding the 2007-09 panel). Do the following plots with year in the x axis and the variable I describe below in the y axis:

Plot 1 mean and median income (adjust by inflation).
Plot 2 Gini coefficient income.
Plot 3 mean and median labor earning (adjust by inflation).
Plot 4 Gini coefficient labor earning.
Plot 5 mean and median financial assets (adjust by inflation).
Plot 6 Gini coefficient financial assets.
Plot 7 mean and median housing value (adjust by inflation).
Plot 8 Gini coefficient housing value.
Plot 9 mean and median mortgage debt (adjust by inflation).
Plot 10 Gini coefficient mortgage debt.
Plot 11 mean and median credit card debt (adjust by inflation).
Plot 12 Gini coefficient credit card debt.